Exos Aerospace Systems & Technologies, Inc. | Project GUNGNIR

AF TACFI | SBIR Phase II FA9300-25-C-6022

Refuel · Relight · Reuse: In-Space Refueling + Storable-Propellant RDRE Flight Demo for Fast Transfer Vehicles (FTV/OTV)

Focus: Space Mobility & Logistics / in-space propulsion



BLK3 reusable VTVL vehicle — tethered hover test, Greenville TX

Concept of Operations

1 **Hover**

2 **Flight to 100 km+**

3 **In-space: RDRE fire + restart & µg propellant transfer**

4 **Land · recover · refly**

Space logistics needs FTVs that can be refueled and re-maneuvered on orbit. Exos flies the refueling and propulsion hardware in space, recovers it, and flies it again.

Objective

• Fly a **non-cryogenic (storable) RDRE** on the reusable BLK3 vehicle: hover → flights to 100 km+ → **in-space ignition, restart and pulsing**.

• Same campaign: **tank-to-tank storable-propellant transfer in microgravity**, the refueling function FTVs and depots need.

Innovation

• Recoverable **in-space** test of refueling + detonation propulsion: vacuum ignition/restart and µg feed/transfer can't be done on a ground stand.

• **Beyond HVRR/Hermes** (atmospheric VTVL hops): GUNGNIR takes the RDRE to vacuum, coupled to µg refueling. Reusable = repeated in-space test points.

Operational Impact

• Refuelable FTVs = **maneuver without regret** for USSF mobility & logistics; refuel and service on orbit instead of reentry disposal; stations as **refueling depots**.

Deliverables

• 1 hover + 2 flights to 100 km+ (in-space RDRE firings + µg transfer); flight dataset (thrust, Isp, restart, transfer) + final report.

End User: AFRL/RJPE (Edwards) · **Customer:** AFRL — Dr. M. Harvazinski; J. Burr (RDRE lead)

Engine: government-furnished / candidate storable RDRE (GTL-built), selected by program trade study. Tanks: Type V composite heritage (Scorpius, flew on IM Odysseus & Athena).

Integrated Cost and Schedule ($M)

$M	Year 1	Year 2	Total
Dev: integration, storable feed + transfer module	1.4	0.3	1.7
Prototype: hover test	0.6	—	0.6
Test flights: in-space fire/restart + µg transfer	—	1.7	1.7
TACFI (SBIR/federal)	1.0	1.0	2.0
Match: Texas Space Commission (SEARF)*	1.0	1.0	2.0
Total	**2.0**	**2.0**	**4.0**

Milestones: M1 PDR (Q1) · M2 CDR (Q2) · M3 Hover (Q4) · M4 Flight 1: in-space RDRE fire (Q5) · M5 µg transfer demo (Q6) · M6 Flight 2: restart/pulse + final report (Q8)

*TSC board meets 11 Dec 2026; match subject to board approval and TACFI selection. Year splits are planning estimates.

Success & Transition (DoD)

Success: storable-RDRE ignition and restart in space, plus a µg propellant transfer, both recovered and reflown, with the flight dataset delivered to AFRL. No NASA dollars required; the path is DoD-only.

1 **Now: SSC SML In-Domain Orbital Logistics Challenge** — concept paper on propellant management, tank refuelability and FTV integration (Q4 CY26) — **share of $20M**

2 **TACFI on FA9300-25-C-6022** — resumes when guidance issues; sponsor AFRL/RQ Rocket Propulsion — **$2M + $2M TSC**

3 **Sole-source SBIR Phase III (15 USC 638(r))** — AFRL/RQ + RV, bridged by the SPRINT BAA — **$2–10M**

4 **STRATFI Government-Only ($1 SBIR : $2 DoD)** — or Strategic Breakthrough (≥20% DoW); SML as match signatory — **$9–45M**

5 **SpEC OTA prototype → program of record** — SAML PE 1203955SF, RG-XX refuel suppliers, 'high-thrust reusable OTV' (2030–35) — **$10–60M**

Distribution: non-proprietary summary. TACFI is subject to selection; nothing is awarded. Exos provides the reusable flight-test platform, integration and propellant-transfer hardware; RDRE hardware is government-furnished or selected by trade study.

GUNGNIR — DoD Transition Path

DoD-only ladder, no NASA share required. Refueling/logistics leads; storable RDRE is the FTV propulsion enabler.

Why refueling leads

• USSF Objective Force plan (Apr 2026): demonstrate refueling + augmented-maneuver tugs 2025–30; make refueling operational + **high-thrust reusable OTVs** 2030–35; on-orbit logistics architecture 2035–40.

• USSPACECOM Dynamic Space Operations / **maneuver without regret** needs on-orbit resupply. RG-XX is the first NSS program with a refueling requirement ($355M FY27; >$2.7B over the FYDP).

• Commercial stations as **refueling depots**: a second revenue line that helps make LEO stations sustainable. SSC's 'orbital warehousing' topic is the closest DoD demand signal today.

Sustainment / environment

• NOAA WB-57 sampling found spacecraft-reentry metals (Al, Cu, Li, **niobium**, hafnium) in ~10% of stratospheric aerosol particles (Murphy et al., PNAS 2023).

• Models project mega-constellation reentries could exceed 360 t/yr of alumina, with open questions on ozone and upper-atmosphere effects (GRL 2024; JGR 2025).

• Refuel, service and reuse on orbit lower the number of vehicles disposed of by reentry, which fits debris rules and the atmospheric concern alike.

Exos role

• Reusable VTVL flight testbed to 100 km+ (in-space firing + µg transfer, recovered and reflown), integration, and Type V tank / zero-loss transfer systems.

• Engine: storable RDRE, government-furnished or candidate GTL build, selected by trade study. Interfaces designed to mate with SERB-approved refueling ports (RAFTI / PRM).

1 · Now: SSC SML In-Domain Orbital Logistics Challenge ($20M)

• SpaceWERX + SSC Servicing, Mobility & Logistics. Topics: propellant management, tank refuelability, transfer metering, FTV/OTV. Also: SSC Front Door intake + SpEC (NSTXL) membership.

2 · TACFI on FA9300-25-C-6022

• On hold pending AFWERX/SpaceWERX guidance. Submit within the next window. Sponsor: AFRL/RQ Rocket Propulsion (GS-15 branch/division lead). Match: TX Space Commission SEARF $2M (board 11 Dec 2026).

3 · Sole-source SBIR Phase III

• 15 USC 638(r) off the Phase II; no competition required. Non-SBIR funds from AFRL/RQ (PE 0602203F / USSF PE 1206601SF, Rocket Propulsion Technology) and AFRL/RV for the in-space mission. SPRINT BAA white paper as the S&T bridge.

4 · STRATFI, DoD-only

• Government-Only Matching Option: $1 SBIR : $2 non-SBIR federal (e.g., $3M + $6M), no private or NASA share. Alternative: Strategic Breakthrough (up to $30M, ≥20% DoW program funds). Match signatory: SSC SML.

5 · OTA prototype → program of record

• SpEC OTA prototype ($10–60M), then SAML PE 1203955SF, supplier to RG-XX refueling vendors, and the Objective Force high-thrust reusable OTV line (2030–35).

Ask of the GS-15

• (1) Sponsor TACFI when guidance posts. (2) Name the Phase III funding office. (3) Introduce us to SSC SML for the Orbital Logistics Challenge and the STRATFI match.

Sources: USSF Objective Force (Breaking Defense, Apr 2026); SSC SML / SpaceWERX Orbital Logistics Challenge (SSC, May 2026); AFWERX PY26.1 STRATFI Matching Funds Guidance; Murphy et al., PNAS 2023. TACFI is subject to selection; nothing is awarded. Program names and $ ranges are planning estimates.